

Experiment 1: Exploring AWS CloudShell and the AWS Cloud9 IDE

Both tools are provided by Amazon Web Services and allow developers to work with AWS services directly from a web browser without installing software on their computer.

AWS CloudShell

AWS CloudShell is a **browser-based command line interface (CLI)** environment that allows users to run AWS commands directly from the AWS Management Console.

Features

Ready-to-use **Linux shell environment**

AWS CLI, Python, Node.js, Git already installed

Persistent storage in the home directory

Accessible directly from **AWS Management Console**

No setup or configuration required

Uses

Running **AWS CLI commands**

Managing AWS services such as **S3 buckets, EC2 instances, DynamoDB**

Automation and scripting

Deploying and testing applications quickly

2. AWS Cloud9

AWS Cloud9 is a **cloud-based Integrated Development Environment (IDE)** used for writing, running, and debugging code.

Features

Online **code editor**

Supports **multiple programming languages** (Python, JavaScript, Java, etc.)

Built-in **terminal**

Real-time **collaboration with other developers**

Direct integration with AWS services

Uses

Writing and debugging **applications**

Developing **AWS Lambda functions**

Building and testing **cloud applications**

Collaborative coding

Working with AWS CloudShell Interface

Step 1: Login to AWS Console

Open: <https://aws.amazon.com>

Click Sign In → Console

Enter credentials

Step 2: Open CloudShell

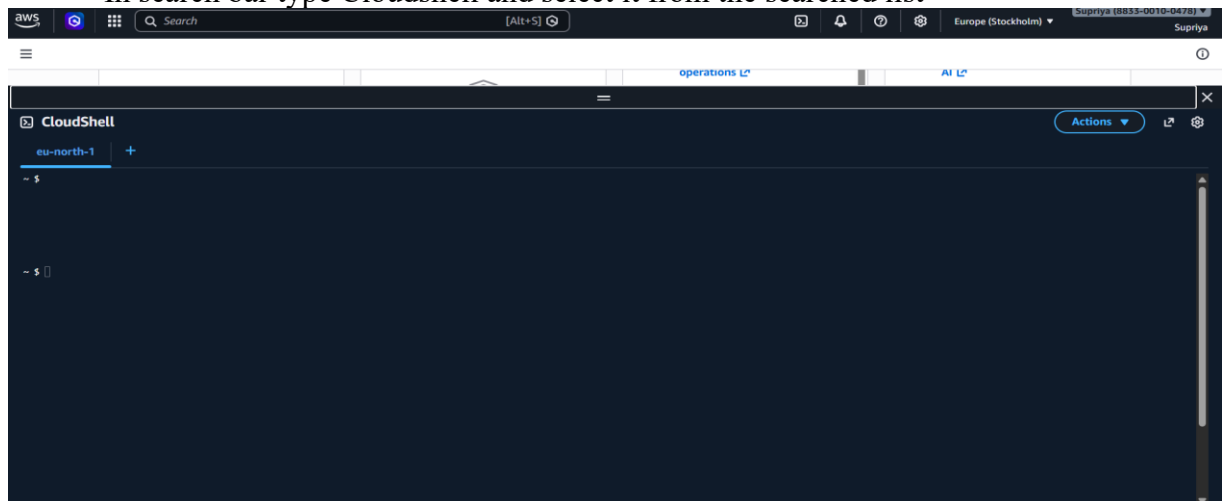
Locate the CloudShell icon (terminal icon) on top right

Click it

Wait for environment to start (first time takes ~1 minute)

Or

In search bar type Cloudshell and select it from the searched list



Step 3: Run Basic Commands

1. Check Version:

`$aws --version`

2. List S3 buckets:

`$aws s3 ls`

3. Create a bucket (replace name):

`$aws s3 mb s3://my-unique-bucket-name`

4. Create a simple text file:

`$echo "this is first cc lab Program in CSE dept of SKIT" > test.txt`

or

`nano test.txt`

Type the contents of file

5. View the content of file

\$cat test.txt

6. View contents of bucket

\$aws s3 ls s3://bucket-name

7. Upload file:

\$aws s3 cp test.txt s3://my-unique-bucket-name

8. View contents of folder in bucket

\$aws s3 ls s3://my-bucket/folder-name/

9. View recursive listing in bucket

\$aws s3 ls s3://my-bucket --recursive

10. Delete a Single File

\$aws s3 rm s3://bucket-name/file-name

11. Delete All Files in a Bucket (Empty Bucket)

\$aws s3 rm s3://bucket-name --recursive

12. Delete All Files in a Bucket (Empty Bucket)

\$aws s3 rm s3://bucket-name/folder-name/ --recursive

13. Delete an Empty Bucket

\$aws s3 rb s3://bucket-name

14. Delete Bucket with All Contents (Force Delete)

\$aws s3 rb s3://bucket-name --force

```
cloudshell-user@ip-10-0-170-32 ~ $ aws --version
aws-cli/2.4.10 Python/3.8.8 Linux/5.10.84-99.471.amzn2.x86_64 exe/x86_64.amzn.2 prompt/off

aws s3 ls
2024-04-23 15:27:28 my-unique-bucket-name

aws s3 mb s3://my-unique-bucket-name
make_bucket: s3://my-unique-bucket-name

echo "this is first cc lab Program in CSE dept of SKIT" > test.txt

cat test.txt
this is first cc lab Program in CSE dept of SKIT

aws s3 cp test.txt s3://my-unique-bucket-name
upload: ./test.txt to s3://my-unique-bucket-name/test.txt

aws s3 ls s3://my-unique-bucket-name
2024-04-23 15:28:42 test.txt

aws s3 rm s3://my-unique-bucket-name/test.txt
delete: s3://my-unique-bucket-name/test.txt

aws s3 rb s3://my-unique-bucket-name --force
remove_bucket: s3://my-unique-bucket-name

cloudshell-user@ip-10-0-170-32 ~ $
```